

Altamash Khan

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AI Engineer who builds production **LLM agents for WhatsApp commerce** end to end – **LangGraph** orchestration, catalog-constrained **structured outputs**, **RAG**, and the **evaluation harness** that keeps them honest. Top contributor to a multi-bot Python/FastAPI platform with **1,400+ commits in 11 months**, owning the intent classifier, product parser, cart engine and B2C bot. Comfortable anywhere from an OpenAI schema limit to a Redis Lua lock.

EDUCATION

Jamia Hamdard

B.Tech in Computer Science — New Delhi

CGPA: 8.6

Sept 2023 – May 2026

SKILLS

Generative AI: OpenAI API, LangGraph, LangChain, Multi-Agent Systems, Tool Calling, Structured Outputs / JSON Schema, RAG, Embeddings, Prompt Engineering, Intent & Entity Classification, CLIP

LLM Evaluation: LLM-as-Judge, Synthetic Scenario Generation, Deterministic Validation, Regression Testing, HTTPX Interception, Backend Mocking

Backend & APIs: Python, FastAPI, Async/asyncio, Pydantic v2, Node.js, Express.js, REST APIs, WhatsApp Business API, Webhooks

Databases, Search & Infra: PostgreSQL, asyncpg, Redis, Pinecone, Elasticsearch (kNN), MongoDB, MySQL, Docker, Kubernetes, Cloudflare R2, Git

EXPERIENCE

Multiply Tech India Pvt. Ltd.

Noida, UP

Nov 2025 – Present

AI Engineer

Nov 2025 – Present

- Architected the **LangGraph WhatsApp ordering agent** – a 7-node StateGraph with 4 conditional routers over a hierarchical structured-output intent taxonomy of **18 entities × 51 operations × 63 sub-operations**, dispatched to 90+ handlers. Primary author of the 2,300-line classifier and 4,000-line cart and product executors.
- Built a **catalog-constrained Product Parser** that maps free-text Hinglish and photographed order slips onto each seller's live catalog. Beat OpenAI's ~990-value enum cap (one catalog dimension had **2,611 designs**) with first-fit-decreasing enum sharding – up to **24 parallel LLM batches**, merged back into one parse.
- Scaled seller-context prompt generation to a **38.8k-item catalogue** that produced 831k tokens and overflowed the model input – capped the catalogue block at 250k tokens and parameter summaries at 300 values per dimension, deleting 1,172 lines.
- Built **LLM evaluation and regression infrastructure** – catalog-grounded multi-turn scenario generation over a code-derived taxonomy of **120 intents** (95 handled, 19 capabilities in 3 difficulty tiers), deterministic cart/intent/quantity checks before an **LLM-as-Judge**, and severity-ranked issue reports in PDF and Postgres.
- Replaced scattered mock flags across ~70 backend clients with one **HTTPX transport interception** layer, isolated per scenario with ContextVars so parallel test runs never leak state – while search and cart stay live.
- Made concurrent chats safe with **per-session turn serialisation on Redis** – SET NX locks with Lua compare-and-extend/release, heartbeats and dead-holder takeover – plus an LLM **queue-join** that folds

rapid follow-ups (“bisleri” then “jeera dikhao”) into a single search reply.

- Shipped a **12-node B2C shopping bot** (+29k lines) with multi-store search and cart, price filters, complaint and human-support flows, and a tool-calling **conversational agent** on gpt-5.2 with 6 tools (orders, outstanding balance, KB RAG, catalog search) capped at 4 tool rounds.
- Built **per-store RAG on Pinecone** – text-embedding-3-large, 750/150-token chunking, one namespace per store, source-cited answers in the buyer’s own language and script (Hindi, Hinglish, English).
- Cut latency and LLM spend – Redis translation cache with fail-open, translate-once chunking (3 round trips → 1 per long reply), `asyncio.gather` fan-out, background webhook processing, Redis-cached DB prompts, cost tracking across 9 modules, and routing across **7 OpenAI model families**.
- Built buyer-facing flows end to end – purchase-history **reorder suggestions** with a store-wide fallback, discount negotiation, agent-call confirmation that holds backend dispatches until the buyer says yes, and salesman ordering.

SecOrigin Technologies

Oct 2024 – Mar 2025

Remote

Software Developer Intern

Oct 2024 – Mar 2025

- Developed and optimised **15+ REST APIs** with Node.js and Express.js, improving response time by **30%** and implementing JWT-based authentication.

PROJECTS

CLIP Image-Match Service

Python, FastAPI, open_clip, Elasticsearch, Postgres, R2

- Standalone microservice matching buyer photos to catalogue SKUs – ViT-B-32 512-dim embeddings in Elasticsearch kNN, multi-store filtering, per-SKU dedup and **confidence tiers by score and margin**; Postgres-backed re-sync re-indexes without re-running inference.

Delhi Police Citizen-Help Bot

Python, FastAPI, Pinecone, OpenAI, WhatsApp API

- WhatsApp bot answering citizens’ legal and procedural questions from policy documents, reusing the RAG pipeline with **source-linked answers** and dedicated handling for officer-refusal scenarios.